M5 Assignment | Exploring Data Literacy, Mining Methods, and Storage Solutions

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AAI101 – Introduction to AI
Due: Tuesday, September 29, 2026

Part 1: Understanding Data Literacy and Its Importance

To me, data literacy is pretty basic: can I read a set of numbers, figure out what they're saying, make a chart that doesn't lie, and explain it to someone who doesn't live in spreadsheets? That skill matters more every week. School, work, and even small money choices lean on data now, and with LLMs everywhere it's easy to trust a clean answer without checking the story underneath. Negangard writes about using big data and analytics in accounting to catch fraud earlier, before losses stack up (Negangard). The habit he describes — notice the weird pattern, ask why, then say it in plain words — is what I'm trying to get better at. Same idea shows up in class with quiz tables or attendance logs, and it shows up in banking when someone flags a charge that doesn't fit. If the rows are solid and I can walk another person through what they mean, the decision is a lot easier to stand behind.To me, datareading data, understanding what it says, creatingexplainingvisualizations. As emerge develop, I needinstead of accepting an AI answer at face value. For example, a school could use attendance and quiz data to notice that a student needs extra help, while a consumer might compare a spending chart before changing a budget. Clear visuals and a good story help people question a large dataset and understand what action makes sense.In finance, that skill also supports fraud prevention: building data literacy helps people use analytics and visualization tools to spot unusual patterns across full sets of transactions, such as repeated small charges or activity in an unexpected place, and catch fraud earlier—before losses spread (Negangard).

[Word count Part 1: 163]

Part 2: Structured vs. Unstructured Data

Structured data sits in tables with rows and columns. Unstructured data doesn't have that fixed layout — think text, images, audio, or video.

Structured Data (5 examples)	Unstructured Data (5 examples)
Student roster: student_id, name, grade level	Teacher comments in an email or chat
Attendance log: date, student_id, present/absent	Classroom photos or scanned worksheets
Quiz score sheet: student_id, quiz_average	Student voice recordings for a project
Inventory list: SKU, quantity, shelf location	Social media posts about a school event
Sales table: order_id, item, units, price	PDF essays or free-text peer reviews

Part 3: Introduction to Data Mining Methods

Method chosen: Classification — each row already has a label (Pass or Needs Help).

No-code tool: Microsoft Power BI Service (https://app.powerbi.com).

Dataset: AAI101_M5_Part3_Classification_Fresh.csv (12 fictional students: absences, homework_turned_in, quiz_average, status).

Report name: AAI101_M5_Fresh_Classification_Report

Learning Power BI (what I practiced)

Power BI Service is Microsoft's web app for uploading a table and making charts without writing code. I signed in, dropped AAI101_M5_Part3_Classification_Fresh.csv into My workspace, and built a report called AAI101_M5_Fresh_Classification_Report. On the canvas I put three visuals: a data table with all the student rows, a clustered column chart of Average quiz_average by status, and a clustered bar chart of Count of student_id by status. I had to switch quiz_average from Sum to Average so the column chart compared group means instead of totals. That little click still mattered — wrong aggregation and the chart tells the wrong story. Classification here just means using a labeled column (status: Pass or Needs Help) and comparing other fields across those labels. Clustering would group rows with no labels yet. Association rules look for items that show up together.

Observations and insights

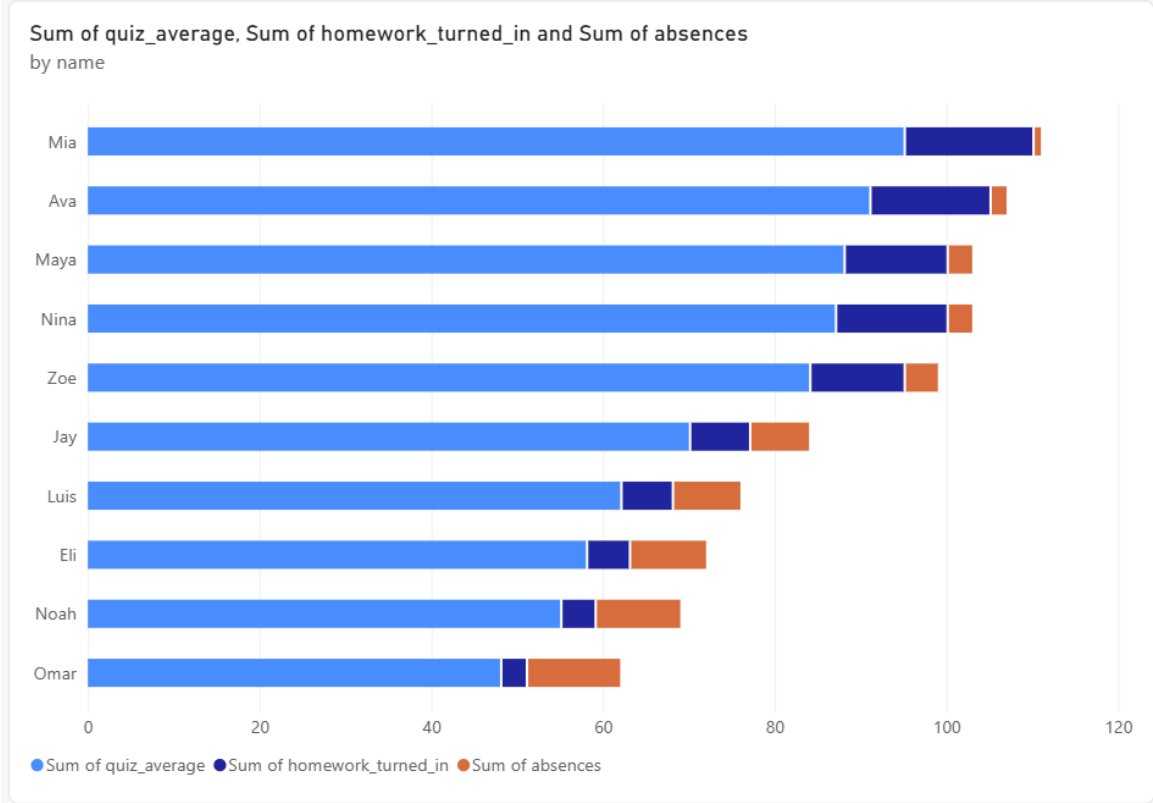
The column chart had a clear gap. Pass students averaged about 91 on quizzes; Needs Help averaged about 55. The bar chart showed 6 Pass and 6 Needs Help, so the gap isn't just from uneven group sizes. In the table, higher absences and fewer homework submissions usually lined up with Needs Help. Labels only help if the numbers behind them are solid. If quiz scores were missing or wrong, the Pass / Needs Help story would fall apart. That loops right back to data literacy and data quality.

Screenshots

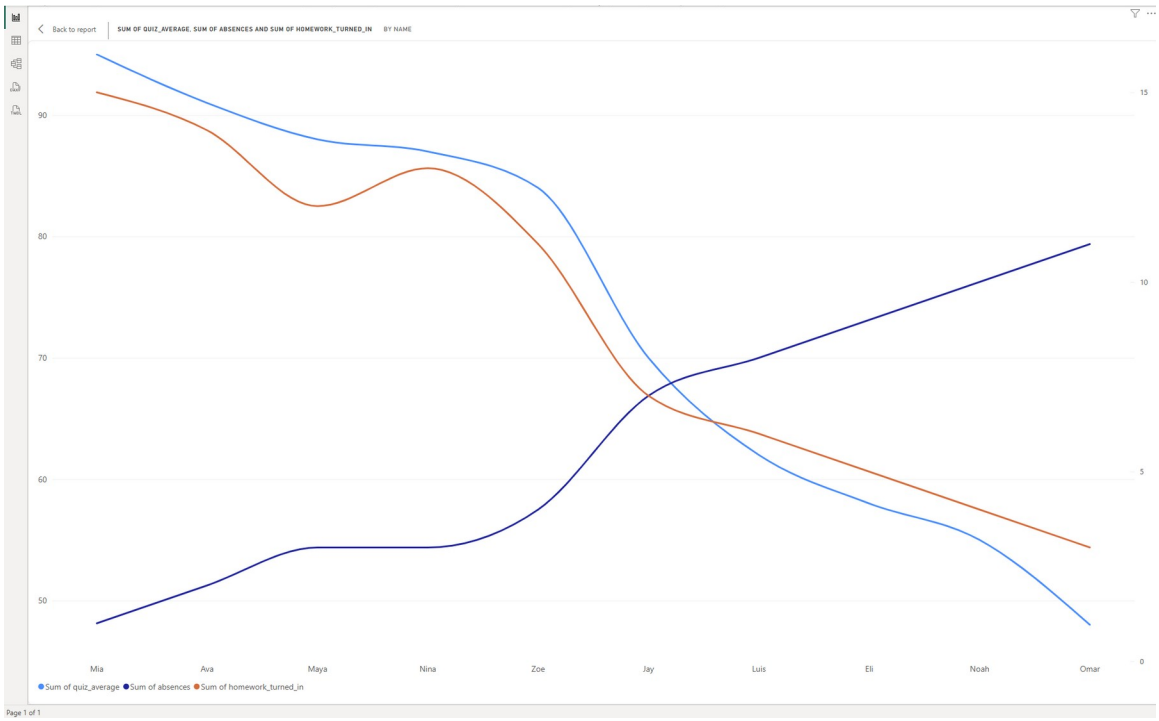
Screenshot 1: Data Table in Excel (What I uploaded to Power BI)

	A	B	C	D	E	F
1	stud	name	absences	homework_turned_in	quiz_average	status
2	S01	Maya	3	12	88	Pass
3	S02	Luis	8	6	62	Needs Help
4	S03	Ava	2	14	91	Pass
5	S04	Noah	10	4	55	Needs Help
6	S05	Zoe	4	11	84	Pass
7	S06	Eli	9	5	58	Needs Help
8	S07	Mia	1	15	95	Pass
9	S08	Jay	7	7	70	Needs Help
10	S09	Nina	3	13	87	Pass
11	S10	Omar	11	3	48	Needs Help

Screenshot 2: Stacked Bar Chart that Lays out Data



Screenshot 3 – Line Chart that arguably shows a point where predictability can happen as absences rise and homework turned in lowers. Student Jay is an example of this.



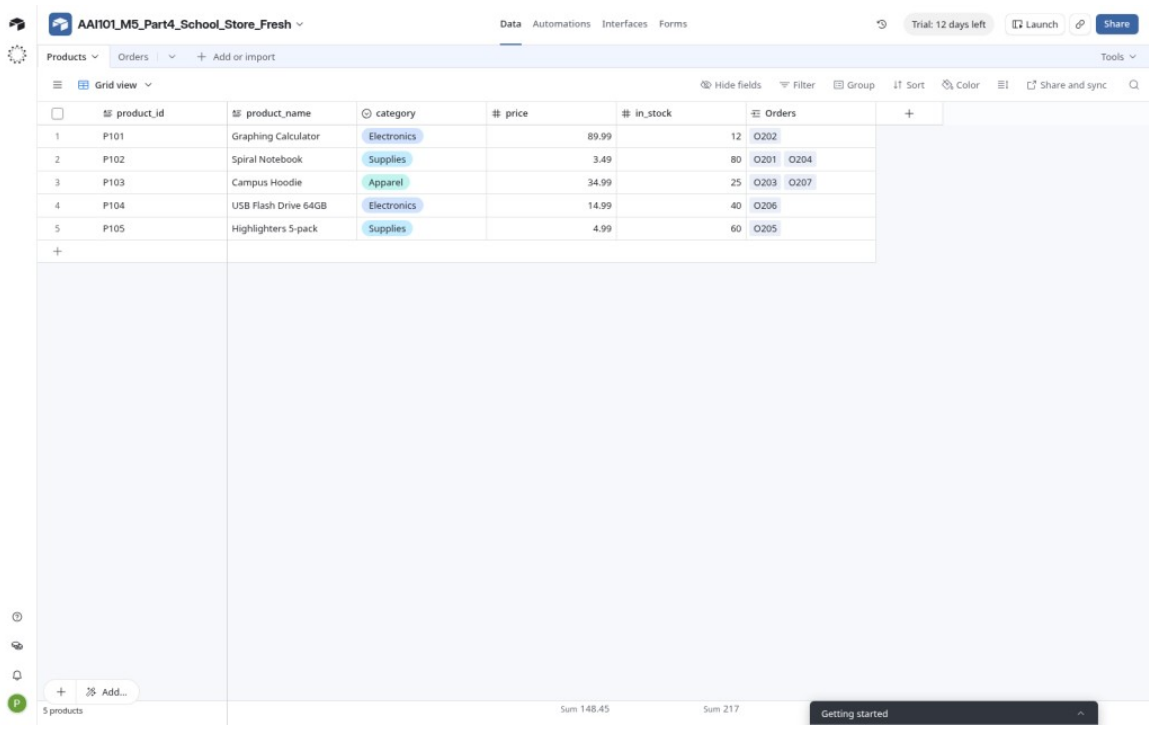
Part 4: Exploring Data Storage Solutions

I used Airtable to build a small school-store base called AAI101_M5_Part4_School_Store_Fresh. It has two tables: Products (what we sell) and Orders (who bought what). Link: https://airtable.com/app6ikKxS0R7TiHAu/tblm2InvnMkpewFmi/viw2hNEg04LY6ywsA?blocks=hide

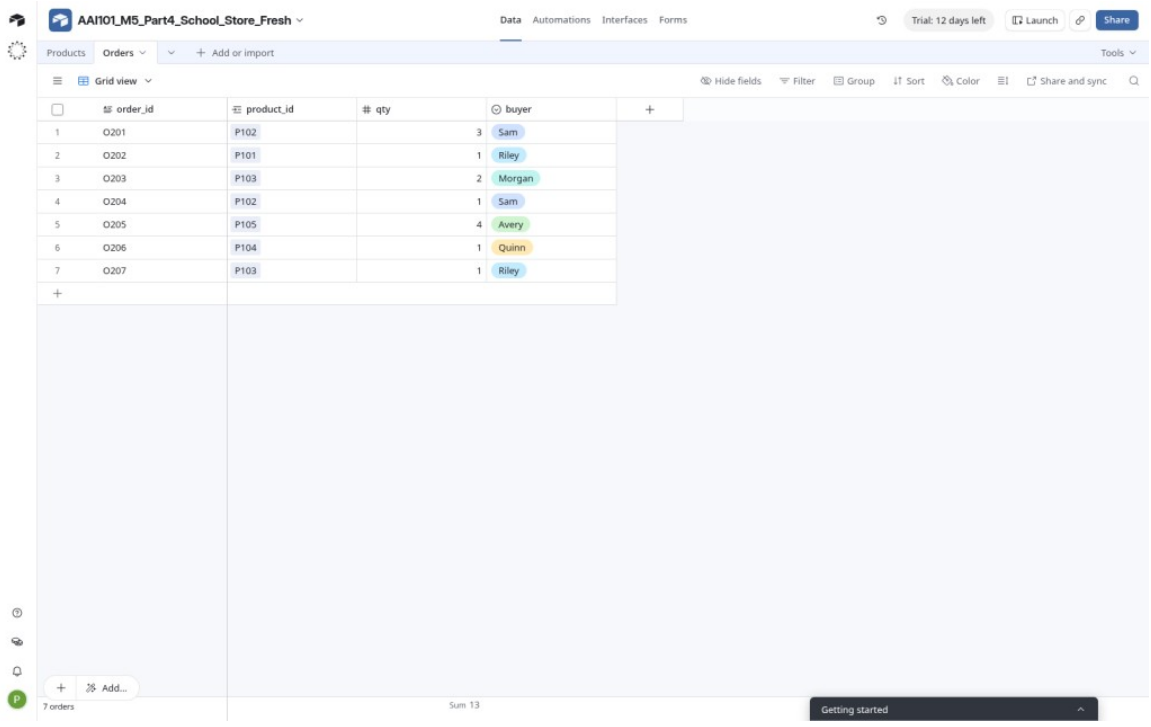
Products has product_id, product_name, category, price, and in_stock (5 products). Orders has order_id, product_id, qty, and buyer (7 orders). The product_id on an order is linked to the Products table, so the tables connect like a simple relational database.

Airtable screenshots

Screenshot: Airtable Products table



Screenshot: Airtable Orders table (linked product_id)



A relational database keeps things in tidy tables that can point at each other, kind of like spreadsheets with links. That's how my Airtable school store works. Products holds each item once — Graphing Calculator, Spiral Notebook, Campus Hoodie, and so on — and Orders just

stores a product_id instead of retyping the name, category, and price on every sale. So if I want to know how many notebook orders Sam placed, I don't dig through duplicate product text. MySQL or PostgreSQL do the same idea at a bigger scale. NoSQL is looser. It doesn't force every record into the same columns, so you could park messy extras like staff notes, photos of damaged stock, or odd buyer details next to the main facts, or even stash a product and its orders in one document. For this school-store set, the linked Airtable tables fit. I'd only reach for NoSQL if I also needed a pile of free-form comments or files that won't sit cleanly in columns.

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Part 5: Reflecting on the Concepts

What stuck with me from this module is that data only helps if people can read it, poke at it, and park it somewhere that matches the job. Classification drops rows into labels you already know — like Pass vs Needs Help in my Power BI charts — so a school can see who might need a nudge. Clustering is for when you don't have labels yet and you're just looking for natural groups, say students with similar absence patterns. Association rules hunt for "these often show up together," which retail uses for marketing and stock. Healthcare leans on similar ideas for risk; schools can use classification for early support. Power BI made the classification piece easy to see with a table and two charts, but the pictures are only as good as the numbers underneath. That's data literacy again. On storage, linked tables like my Airtable Products and Orders (or a normal relational database) are fine for day-to-day reporting. A warehouse makes more sense when you're rolling up years of history from a bunch of systems. NoSQL helps when messy notes, chats, or files need to sit beside the clean numbers. Most places I can think of will mix it: neat tables for the core facts, something flexible for the messy bits, and humans still checking that the story the charts tell is fair.

[Word count Part 5: 225]

Works Cited / Sources

Negangard, Eric. "Stopping Fraud Before It Starts: Big Data and Data Analytics' Potential in Accounting." Experience McIntire, U of Virginia, https://experience.mcintire.virginia.edu/news/big-data-data-analytics-potential-in-accounting/. Accessed 29 Sept. 2026.

MIT Sloan – How to build data literacy in your company: https://mitsloan.mit.edu/ideas-made-to-matter/how-to-build-data-literacy-your-company

Moov AI – Nine Critical Roles for a Successful AI Project: https://moov.ai/en/blog/nine-critical-roles-for-a-successful-ai-project

Microsoft Power BI Service: https://app.powerbi.com

Airtable base (Part 4):
https://airtable.com/app6ikKxS0R7TiHAu/tblm2InvnMkpewFmi/viw2hNEg04LY6ywsA?blocks=hide

AAI101 Module 5 readings/slides on data literacy and unstructured data.